

CUSTOMER CHURN

Exploratory Data Analysis

A plain-language walkthrough of who churns, and why

Dataset: Telco customer records • 7,043 customers • 21 fields

HOW TO READ THIS REPORT

This report is written for readers who may not work with data every day. Each section states one finding, shows the chart behind it, and then explains what it means in plain terms. Technical words (like "chi-square test" or "correlation") are explained the first time they appear and are collected again in the glossary at the end.

1. Executive Summary

Out of 7,043 customers, 26.7% have churned (stopped using the service) and 73.3% are still active. The customers most likely to leave share a recognizable profile: short-term contracts, no add-on support services, and higher monthly bills.

26.7% overall churn rate	39.1% churn on month-to-month plans	47.9% churn on fiber internet	34.9% churn with no tech support
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Bottom line

Contract length is the single strongest signal we found. Customers on month-to-month plans are almost 9x more likely to churn than customers on two-year contracts.

2. About the Data

Each row represents one customer. The columns describe who they are (e.g., senior citizen, has a partner), what services they use (internet type, tech support, online security), how they pay (contract length, payment method, monthly/total charges), and whether they churned.

Item	Value	What this means
Customers (rows)	7,043	One row per customer
Fields (columns)	21	Demographics, services, billing, and the churn outcome
Duplicate rows	0	No repeated customers found
Missing billing values	11	Fixed during cleaning (see Section 3)
Target column	Churn	Yes/No — what we are trying to explain

3. Cleaning the Data

Before analyzing anything, a few small issues in the raw data had to be fixed so the numbers could be trusted.

- 11 customers had a blank "total charges" value (about 0.16% of all customers). These rows were removed rather than guessed at, since the share affected was too small to bias the results.
- Tenure (months as a customer) was grouped into ranges to make patterns easier to see in charts.
- The customer ID column was excluded from analysis — it identifies a customer but carries no information about behavior.
- Yes/No fields (like Churn) were converted to 1/0 so they could be measured and correlated numerically.

Why this matters

Skipping this step is one of the most common mistakes in beginner EDA reports. A single unclean field (like text stored where a number should be) can silently break every chart built from it.

4. The Overall Picture

Before comparing groups, it helps to see the overall split between customers who stayed and customers who left.

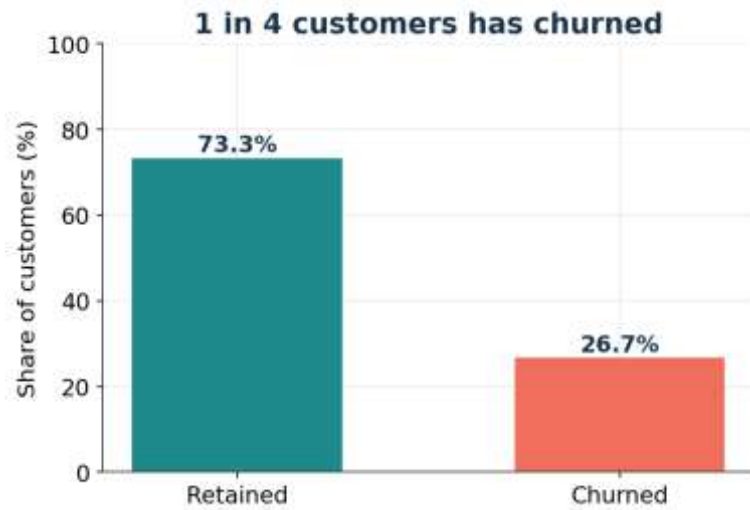


Figure 1 — About 1 in 4 customers in this dataset has churned.

This 73/27 split matters for one technical reason: because far more customers stayed than left, any group comparison later in this report uses percentages (rates) rather than raw counts — otherwise, large groups would look more "churn-prone" simply because they have more people in them, not because they actually churn more often.

5. Contract Type and Payment Method

5.1 Contract length

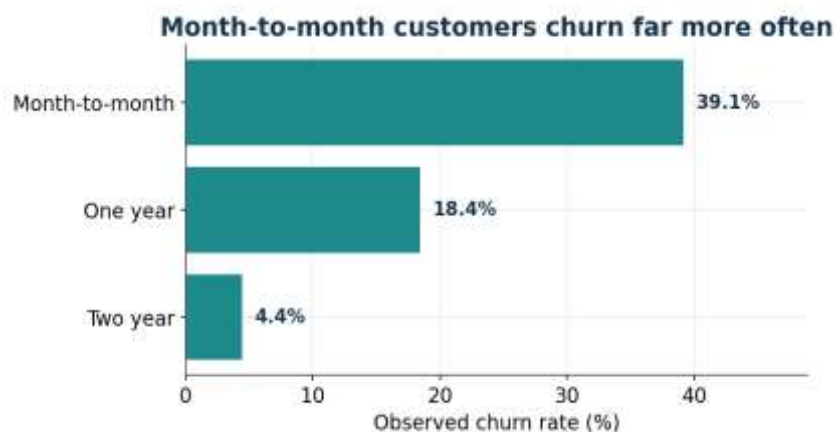


Figure 2 — Observed churn rate by contract type.

Month-to-month customers churn at 39.1%, almost nine times the rate of two-year customers (4.4%). This is the largest gap in the entire dataset.

5.2 Payment method

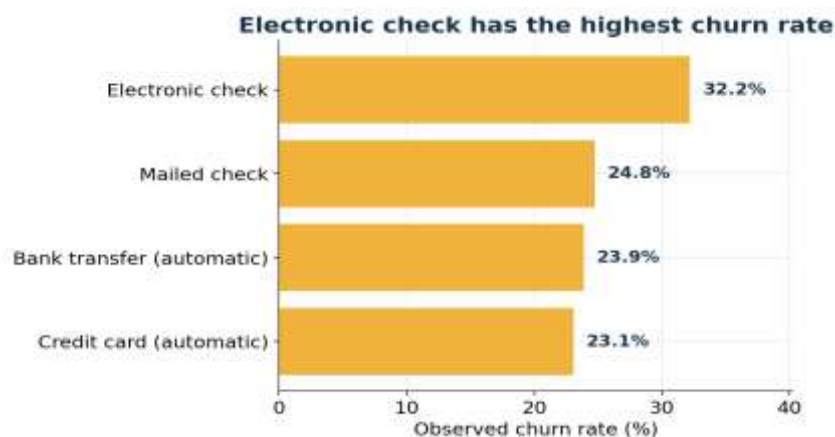


Figure 3 — Observed churn rate by payment method.

Customers who pay by electronic check churn noticeably more often (32.2%) than those on automatic payments (23–25%). This may reflect that manual, non-automatic payment is itself linked to lower ongoing commitment, rather than the payment method causing churn directly.

6. Internet, Support, and Security Services

6.1 Internet service

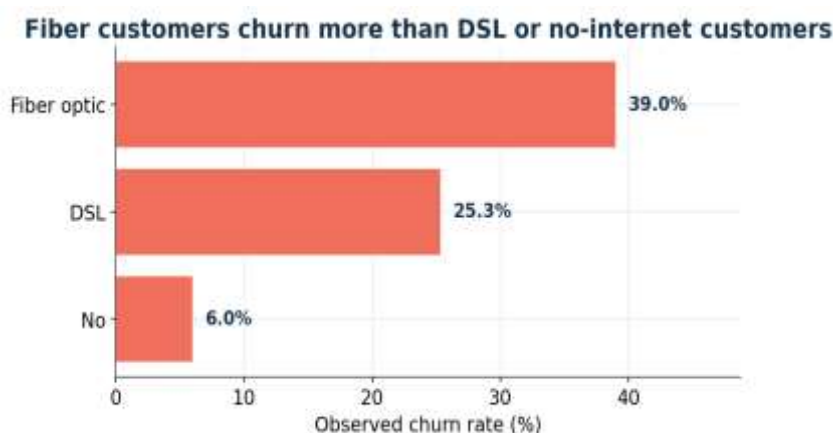


Figure 4 — Observed churn rate by internet service type.

Fiber-optic customers churn the most (39.0%). This is counterintuitive at first — fiber is the faster, more premium service — but it lines up with a pattern shown later: fiber customers also pay the most per month (see Section 8).

6.2 Tech support and online security

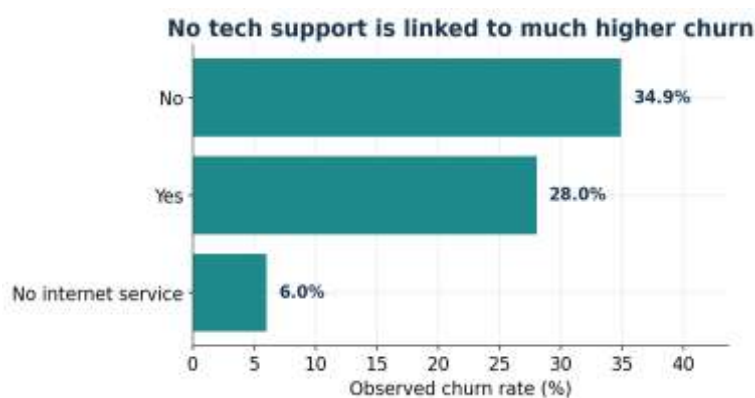


Figure 5 — Observed churn rate by tech support.

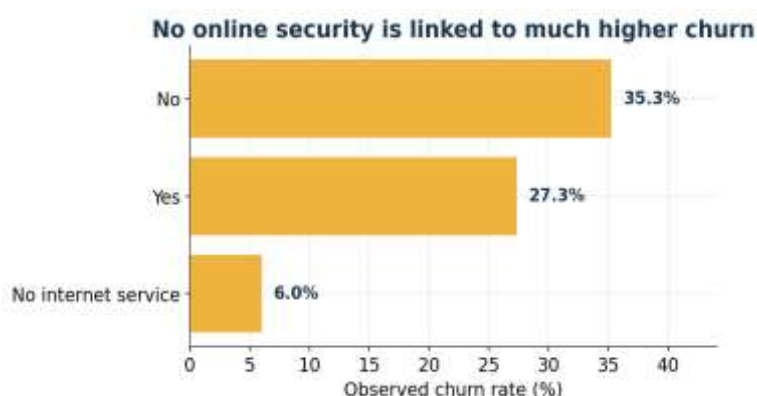


Figure 6 — Observed churn rate by online security.

Both add-on services show the same pattern: customers without the service churn at more than double the rate of customers with it. "No internet service" customers show the lowest churn of all (6.0%), simply because they were never eligible to lose an internet-dependent service in the first place.

7. Customer Segments

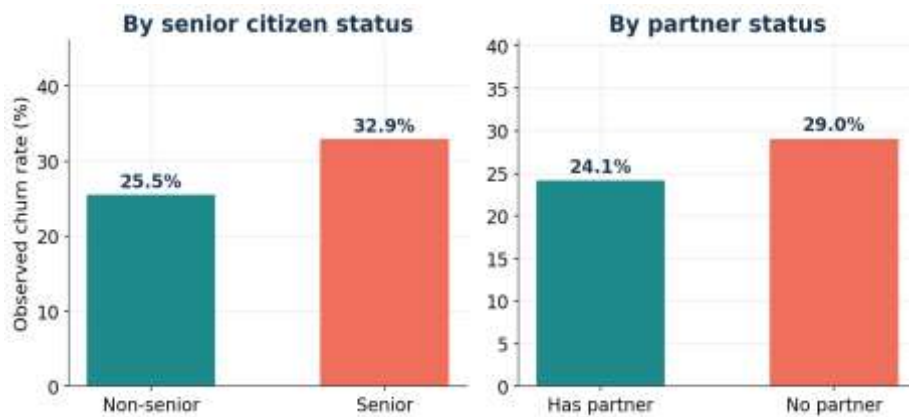


Figure 7 — Observed churn rate by senior citizen and partner status.

Senior citizens churn more than non-seniors (32.9% vs. 25.5%), and customers without a partner churn more than those with one (29.0% vs. 24.1%). Both gaps are real but noticeably smaller than the contract or service-based gaps above — demographics matter less here than how the customer is set up commercially.

8. Tenure and Billing Patterns

Tenure — how many months someone has been a customer — turned out to be one of the most telling signals in the data, so it gets its own section rather than a single line in a bullet list.

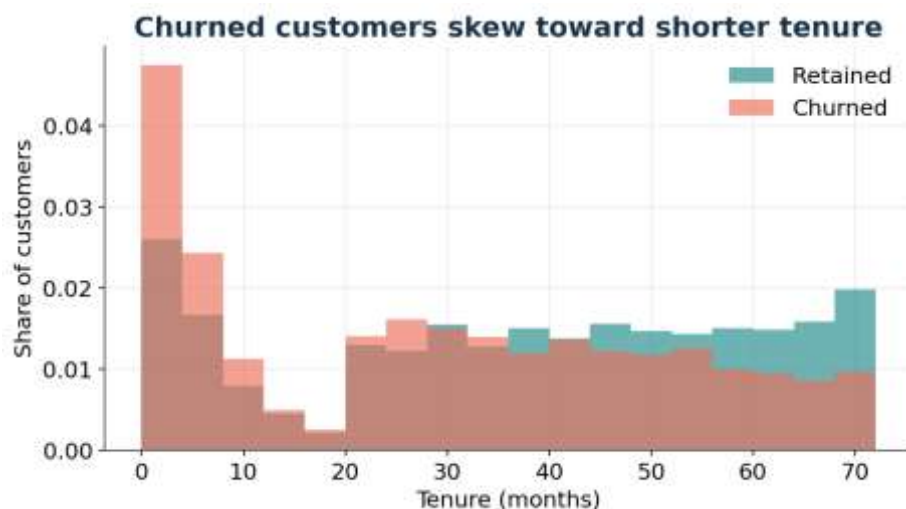


Figure 8 — Tenure distribution for retained vs. churned customers.

Churn is heavily concentrated in the first few months of the relationship: a large share of churned customers leave within their first 0–8 months, while retained customers are spread much more evenly across all tenures, including many long-standing customers past the 60-month mark. On average, churned customers stayed 29.0 months versus 37.3 months for retained customers.

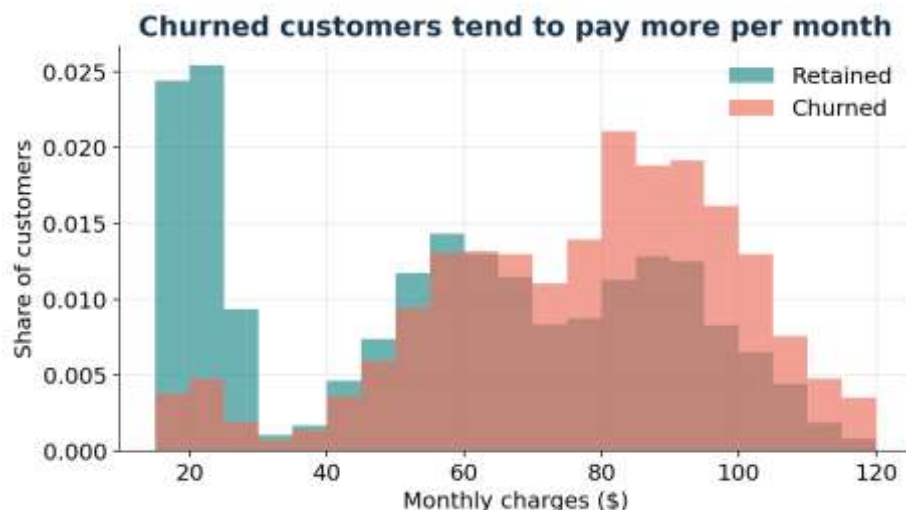


Figure 9 — Monthly charges distribution for retained vs. churned customers.

Churned customers also pay more per month on average (\$76.99 vs. \$58.42). Combined with the tenure chart above, this points to a specific at-risk profile: customers who are both new and on a higher-priced plan.

9. How Reliable Are These Patterns?

A gap between two percentages in a chart could, in principle, happen by chance in a random sample. To check this, a statistical significance test called a chi-square test was run for every category comparison in this report. In simple terms: it estimates how likely it would be to see a gap this large if the factor (like contract type) had no real relationship with churn at all.

Factor	Result	Plain-language read
Contract type	$p < 0.001$	Extremely unlikely to be chance
Internet service	$p < 0.001$	Extremely unlikely to be chance
Tech support	$p < 0.001$	Extremely unlikely to be chance
Online security	$p < 0.001$	Extremely unlikely to be chance
Payment method	$p < 0.001$	Extremely unlikely to be chance
Senior citizen	$p < 0.001$	Very unlikely to be chance
Partner status	$p < 0.001$	Very unlikely to be chance

Important distinction

"Statistically significant" only means the pattern is real and not due to random noise — it does NOT mean one factor causes churn on its own. A customer's contract type, payment method, and add-on services are chosen together, so several of these patterns likely overlap rather than acting independently.

10. How the Signals Relate to Each Other

A correlation heatmap shows how strongly pairs of numeric signals move together, on a scale from -1 (perfectly opposite) to +1 (perfectly together). It helps answer a follow-up question the earlier charts can't: are these risk factors independent, or do they cluster in the same customers?

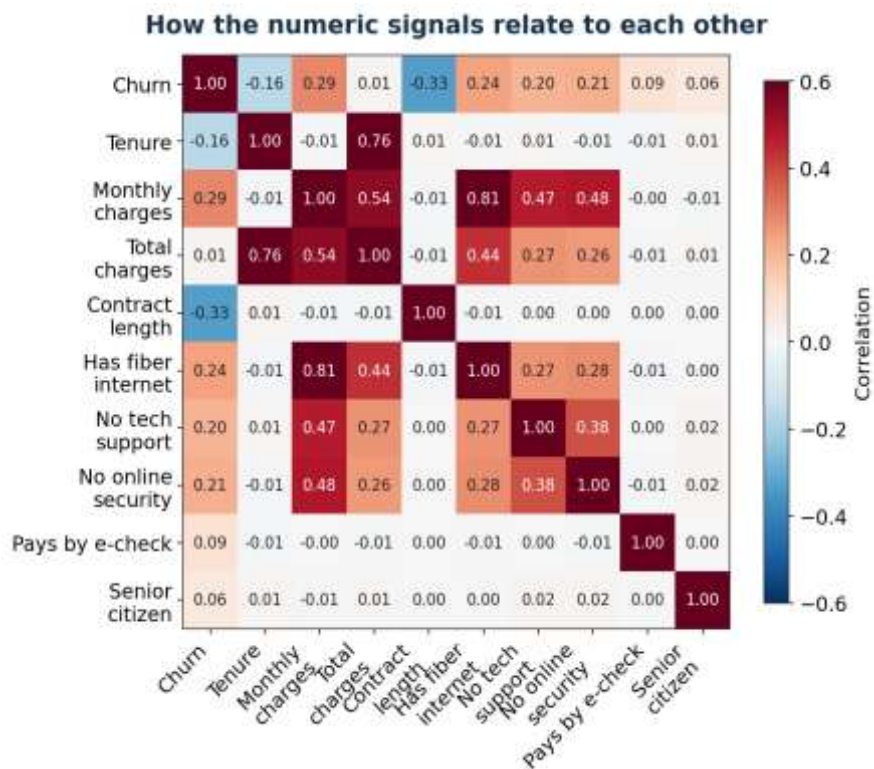


Figure 10 — Correlation matrix of churn and the numeric/encoded fields.

- Contract length has the strongest single relationship with churn (-0.33) — longer contracts, less churn.
- Fiber internet, higher monthly charges, no tech support, and no online security all move together strongly (0.44–0.81 correlations with each other) — this is one bundle of risk factors, not four separate ones.

- Tenure and total charges are highly correlated (0.76), which makes sense: customers who stay longer accumulate more total billing regardless of churn.

In plain terms

There isn't one single cause of churn in this data — there's a cluster of related signals (new + fiber + expensive + no add-ons + short contract) that tend to show up in the same at-risk customers.

11. Key Findings, Summarized

- Overall churn rate: 26.7% of customers have left.
- Contract type is the single biggest factor: month-to-month customers churn 9x more often than two-year customers, and this gap is statistically real, not chance.
- Fiber internet, no tech support, and no online security form a related cluster of risk factors that tend to appear together in the same customers.
- New customers are the most fragile: churn is concentrated in the first several months of tenure.
- Electronic check payers, seniors, and customers without a partner all churn somewhat more — real but smaller effects than contract or service factors.

12. Recommendations

- Prioritize retention efforts on new, month-to-month, fiber customers without tech support or online security — this is the highest-risk combination identified.
- Test whether offering a discounted first-year contract upgrade (month-to-month → one-year) to new customers reduces early churn.
- Investigate the fiber-optic experience specifically (pricing, reliability, or onboarding) since it churns more despite being the premium tier.
- Build a simple dashboard tracking these 4–5 factors together per customer, so at-risk accounts can be flagged automatically rather than found manually.
- Move to a predictive model (e.g., logistic regression) as the next step — this EDA identifies which signals matter; a model can combine them into a single per-customer risk score.

13. Glossary

- Churn — a customer who has stopped using the service.
- Observed churn rate — the percentage of customers within a specific group (e.g., "fiber internet users") who churned. Using rates instead of raw counts allows fair comparison between groups of different sizes.
- Chi-square test — a statistical test that checks whether a difference between groups is likely to be real, or could plausibly have happened by random chance.
- p-value — the output of that test; a very small p-value (e.g., below 0.001) means the pattern is very unlikely to be random chance.
- Correlation — a number from -1 to +1 describing how strongly two measurements move together. Close to 0 means little relationship.
- Tenure — how many months someone has been a customer.

14. Scope and Next Steps

This report covers data understanding, cleaning, and both single-variable and two-variable (bivariate) analysis, plus a first look at how variables relate to each other via correlation. It does not yet include a predictive model — that is a natural next stage once these patterns are validated with the business.

Note on this version of the report — see accompanying message